

OPTIMIZATION PROBLEM FORMULATION

Solar Panel Tilt & Orientation Optimization

A Mathematical Formulation to Maximize Annual Photovoltaic Energy Yield and Efficiency

Engineering Optimization Project | Team Presentation

Real-World Motivation & Context

The Engineering Challenge

Photovoltaic (PV) solar panels generate peak energy when incident solar rays strike the module surface perpendicularly (zero incidence angle).

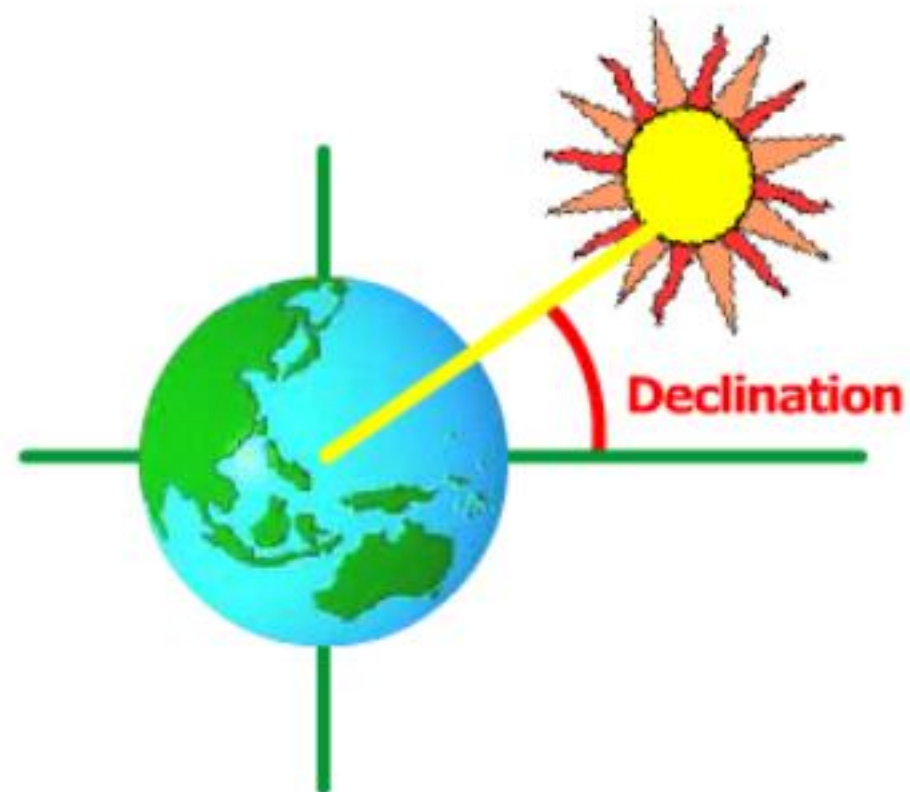
However, the sun's position changes continuously throughout the day (hour angle) and across seasons (declination angle).

Key Decision: Active multi-axis tracking systems are expensive to install and maintain. Determining an *optimal fixed tilt angle* (β) and *azimuth orientation* (γ) eliminates mechanical failure risks while capturing maximum solar irradiance.

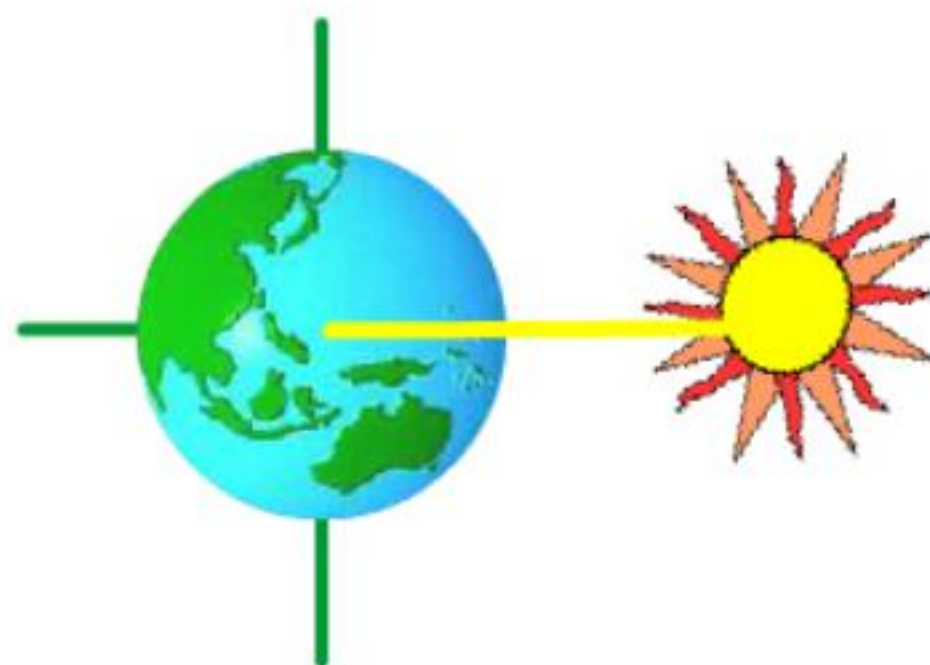


Astronomical & Geometric Variables

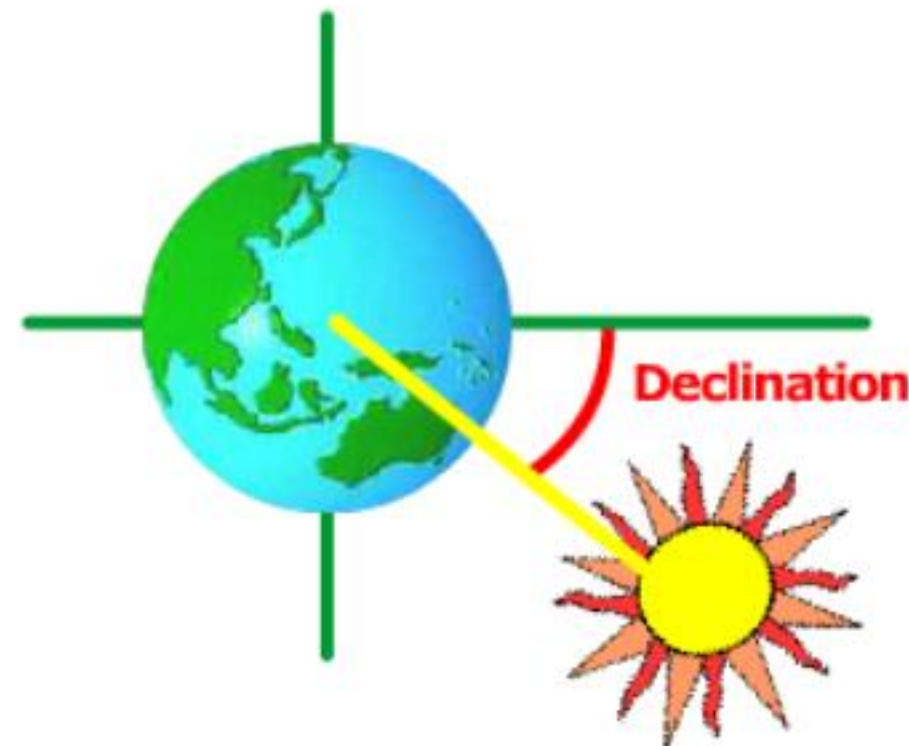
Summer solstice in the northern hemisphere. The declination angle (δ) is at its maximum and is 23.45° .



Spring equinox in the northern hemisphere and autumn equinox in the southern hemisphere. The declination angle (δ) is 0° .



Winter solstice in the northern hemisphere and summer solstice in the southern hemisphere. The declination angle (δ) is -23.45° .



Key Celestial & Position Angles

Solar position is uniquely determined by three deterministic parameters:

Declination Angle (δ):

$$\delta = 23.45 \cdot \sin \frac{360}{365} \cdot (n + 284)$$

Latitude (ϕ): Fixed site location angle north/south of equator.

Hour Angle (ω): Earth's rotation angle (15° per hour from solar noon).

Decision Variables Definition



Tilt Angle (β)

Definition: The angle between the PV module surface and the horizontal ground plane.

Domain: $0 \leq \beta \leq 90^\circ$

Determines seasonal light capture efficiency (steeper for winter, flatter for summer).



Azimuth Angle (γ)

Definition: The panel's surface orientation relative to true South (Northern Hemisphere).

Domain: $-90^\circ \leq \gamma \leq 90^\circ$

Aligns peak daily production with solar noon or afternoon grid demand peaks.



Adjustment Frequency

Fixed Tilt: Single static β year-round.

Seasonal Tilt: Discrete binary decision variables for seasonal adjustments (e.g. $\beta_{\text{winter}}, \beta_{\text{summer}}$).

Objective Function Formulation

Maximizing Annual Solar Energy Yield

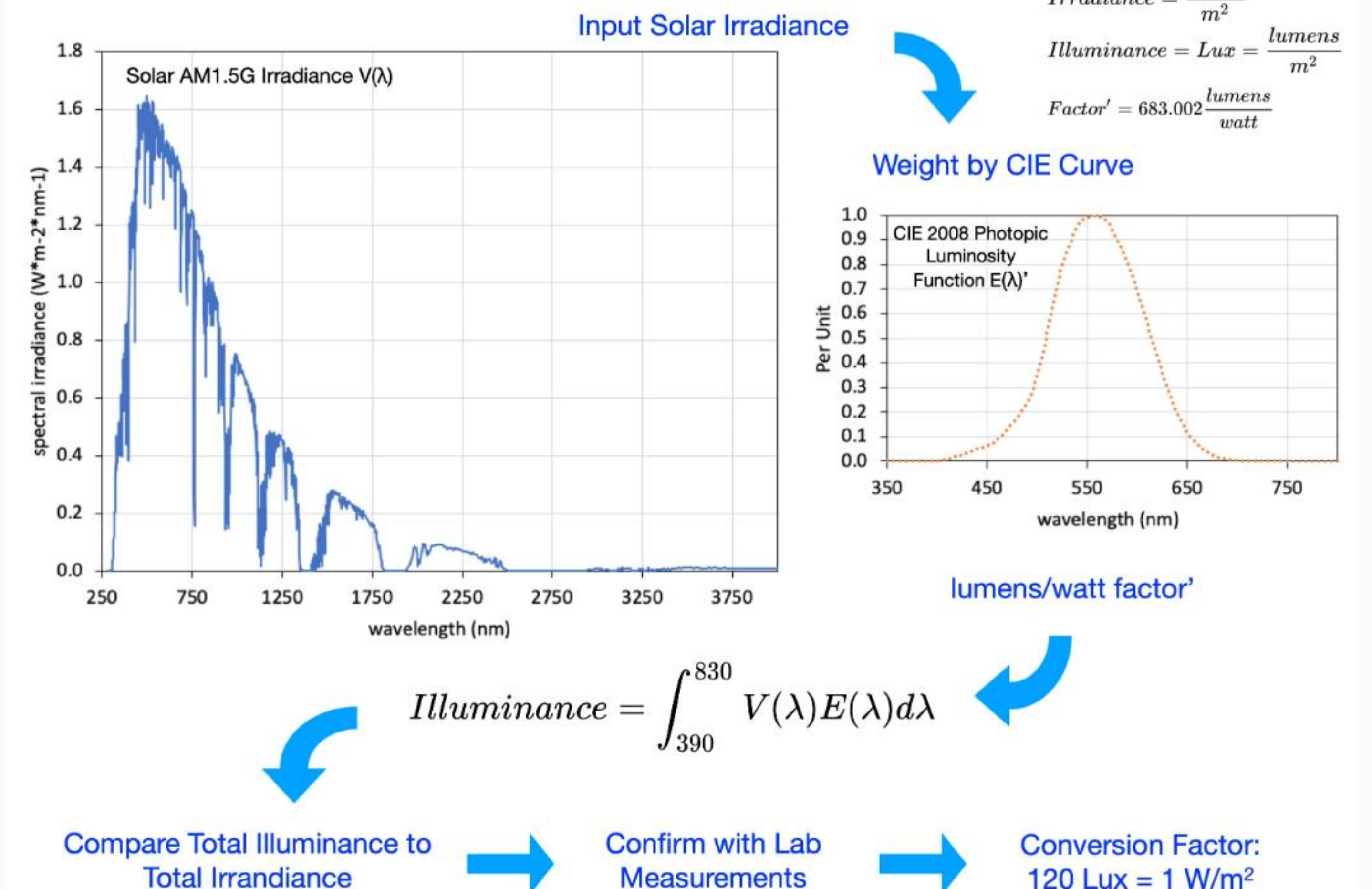
The goal is to maximize the total incident solar radiation collected over a full operational year ($N = 365$ days):

$$\max_{\beta, \gamma} \sum_{n=1}^{365} \sum_{t=\omega}^{\omega} I_T(n, t, \beta, \gamma)$$

Where total irradiance on tilted surface I_T is calculated via Liu-Jordan isotropic model:

$$I_T = I_b R_b + I_d \frac{1 + \cos\beta}{2} + I \cdot \rho \frac{1 - \cos\beta}{2}$$

Conversion: Solar Irradiance to Lux Illuminance



Complete Set of Constraints

Constraint Category	Mathematical Form	Physical & Operational Meaning
Geometric Angle Bounds	$0 \leq \beta \leq 90^\circ$ $-90^\circ \leq \gamma \leq 90^\circ$	Prevents panel facing ground or upside-down orientation.
Self-Cleaning Threshold	$\beta \geq \beta_{\min} \approx 10^\circ$	Ensures rainwater runoff clears dust & debris off glass.
Wind Load / Structural Limit	$\beta \leq \beta_{\max} \approx 60^\circ$	Mitigates extreme overturning wind loads on mounting racks.
Incidence Angle Threshold	$\cos \theta (n, t, \beta, \gamma) \geq 0$	Disregards sunlight striking the backside of the PV panel.

Optimization Classification



Class: Nonconvex NLP

This is a **Continuous Non-Linear Program (NLP)**. The objective function involves trigonometric compositions of $\cos \theta$ and non-linear geometric projection factors.



Convexity Analysis

The objective function $I_T(\beta, \gamma)$ is *non-convex* across the full domain, exhibiting multiple local inflection regions under complex diffuse radiation models.



Solvability Properties

Because the variable space is low-dimensional (2), global optimum can be efficiently found using multi-start gradient methods or direct grid searches.

Dust Accumulation Trade-Offs



Environmental Soiling Penalty Factor

Near equatorial regions, the optical optimal tilt angle β_{opt} is close to 0° (horizontal).

However, near-horizontal panels ($\beta < 10^\circ$) suffer severe performance degradation due to dust accumulation and lack of self-cleaning during rainfall.

Soiling Derate Factor Formula:

$$\eta_{\text{soiling}}(\beta) = 1 - \alpha \cdot \exp(-k \cdot \beta)$$

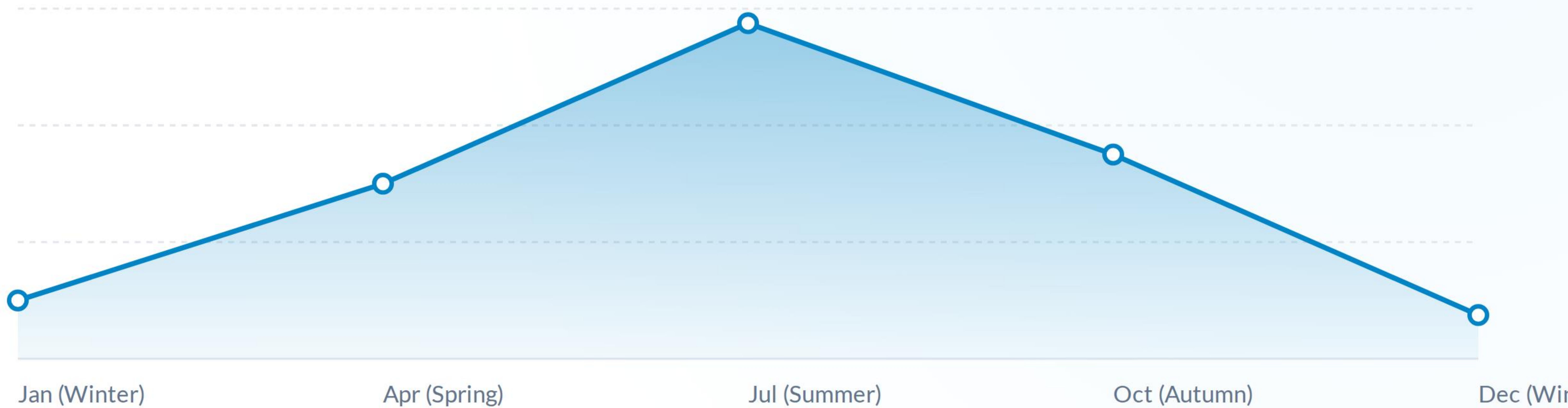
The minimum tilt constraint $\beta \geq 10^\circ$ acts as an essential practical regularization bound.

Case Study Results: Phoenix, AZ



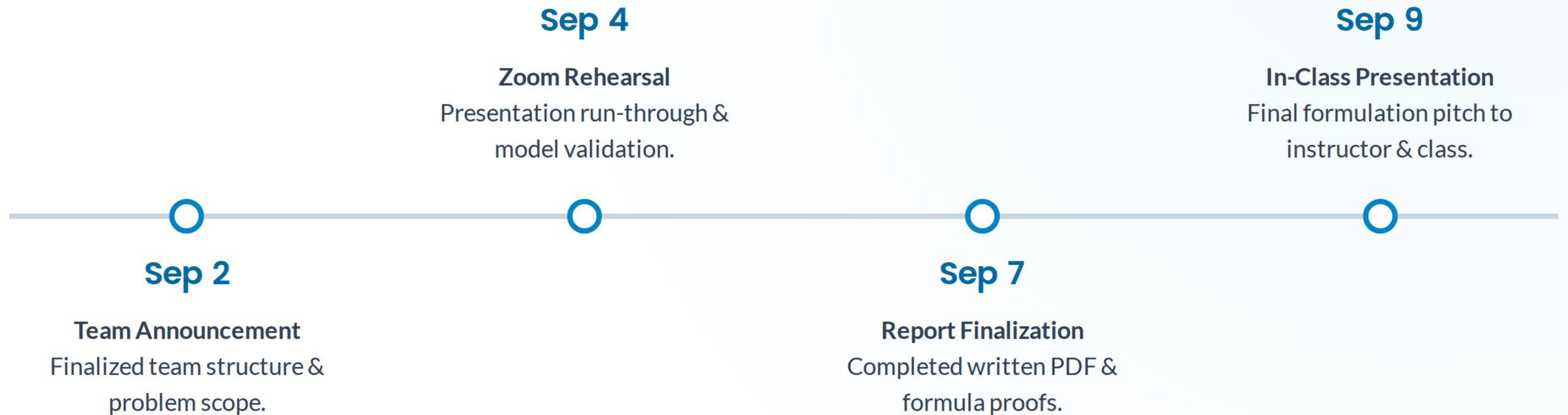
Optimization achieves ~93% of theoretical dual-axis tracking yield at zero additional capital or maintenance cost!

Seasonal Irradiance Trends



Optimized winter tilt ($\beta \approx \varphi + 15^\circ$) flattens seasonal power dips and reduces energy storage battery sizing requirements.

Project Execution Timeline



Questions & Discussion

Thank you for your attention! We welcome any feedback on our mathematical formulation and optimization methodology.

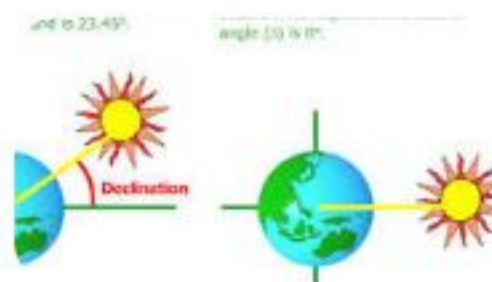
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Image Sources



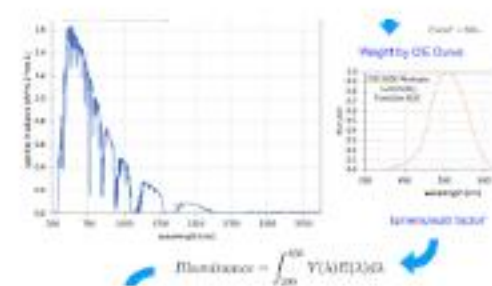
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